Arista Access Node JAY-ACC-55 Failure Resolution SOP (ALARM_8947)

Node: JAY-ACC-55 | **Vendor:** Arista | **Severity:** Critical | **Alarm Code:** ALARM_8947

Description

The occurrence of ASIC memory parity errors, line card crashes, and power supply redundancy loss on Arista's Access Node JAY-ACC-55 (Role: Access) at Jayanagar Metro Site can be attributed to the inherent complexity of Arista's Extensible Operating System (EOS) on the given hardware platform. The EOS, designed to provide high-speed switching and routing capabilities, relies heavily on the efficient management of ASIC memory to perform packet forwarding and switching operations. However, the likelihood of hardware-related issues such as memory parity errors or line card crashes cannot be entirely ruled out due to the possibility of manufacturing defects, environmental stress, or hardware aging. Furthermore, the reliance on power supply redundancy in modern network equipment can lead to unexpected failures when the redundant power supply unit (PSU) fails to compensate for the loss of the primary PSU, resulting in a Critical alarm (ALARM_8947).

Pre-requisite Checks

- Check: Ensure remote hands are dispatched to the Jayanagar Metro Site for physical access and assistance.

- Check: Verify the status of redundant links and connections to prevent service disruption during troubleshooting.

- Check: Confirm the correct CLI dialect (Arista) and EOS version are being used for all diagnostic and resolution commands.

Potential Root Causes

Cause	Probability
ASIC memory parity errors due to hardware defects or environmental stress	High
Line card crashes due to software bugs or compatibility issues	Medium
Power supply redundancy loss due to PSU failure or improper configuration	Low

Diagnostic Commands & Expected Outputs

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Command: show system resource, Expected Output: System resource utilization and memory allocation information, Description: This command provides an overview of the system's resource utilization, including memory allocation, CPU usage, and other relevant metrics. It helps identify potential issues related to memory parity errors or line card crashes.

Command: show power redundancy, Expected Output: Power supply redundancy status and configuration information, Description: This
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command reveals the current status of power supply redundancy, including the status of primary and redundant PSUs, and helps diagnose potential issues related to power supply redundancy loss.

Command: show tech-support, Expected Output: Detailed system information, including hardware and software configurations, Description: This command generates a comprehensive report of the system's hardware and software configurations, which can aid in identifying the root cause of the failure.
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Resolution Steps

Step 1: Step: Replace faulty line card or ASIC, Command: no switch all (to disable switching fabrics) followed by 'reboot' to restart the system, Description: If the diagnostic commands indicate a faulty line card or ASIC, this step involves disabling the switching fabrics to prevent further packet forwarding and then rebooting the system to replace the faulty hardware.

Step 2: Step: Reconfigure power supply redundancy, Command: config power redundancy (to configure the PSU status), Description: If the issue is related to power supply redundancy loss, this step involves reconfiguring the PSU status to ensure proper redundancy and prevent future failures.

Step 3: Step: Update software or firmware, Command: upgrade system software or firmware (to update the EOS version or other software components), Description: If software bugs or compatibility issues are suspected, this step involves updating the system software or firmware to the latest version to resolve the issue.

Post-Check Verification

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Command: show system resource, Expected Output: System resource utilization and memory allocation information (showing no errors or issues), Description: After resolving the issue, this command is run to verify that the system resource utilization and memory allocation are within normal limits.

Command: show power redundancy, Expected Output: Power supply redundancy status and configuration information (showing proper redundancy status), Description: This command is run to verify that the power supply redundancy status is properly configured and functioning as expected.

Command: show tech-support, Expected Output: Detailed system information, including hardware and software configurations (showing no issues or errors), Description: This command is run to verify that the system's hardware and software configurations are within normal limits and show no signs of issues or errors.
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Roll-back Procedures

- Step: Revert software or firmware changes, Command: rollback system software or firmware (to revert to previous version), Description: If the fix fails, this step involves reverting the software or firmware changes made during the resolution process to ensure the system's previous state is restored.

- Step: Restore previous configuration, Command: configure replace (to restore the previous configuration file), Description: If the fix fails, this step involves restoring the previous configuration file to ensure the system's previous state is restored.